Submitted by:

Mariyam Kashif

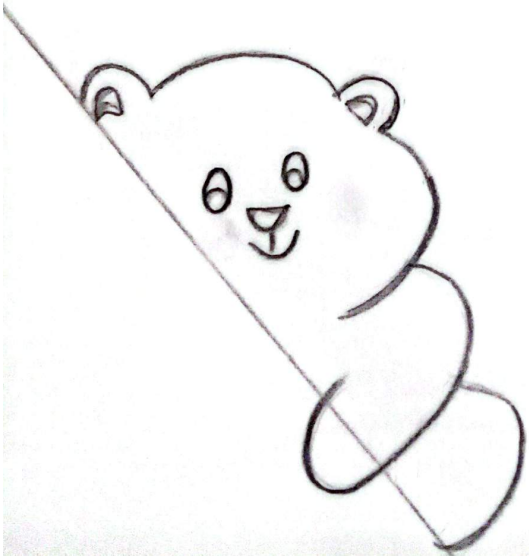
F-25-BSAI-0022

Lab Assignment #01

07-10-2026

Submitted to:

Sir Usama



Assignment #01

Question #01

Code

```
students = []
def result(p,ai,math):
    total = p+ai+math
    per = total/3

    if per >=80:
        grade = "A"
    elif per >=70:
        grade = "B"
    elif per >=60:
        grade = "C"
    elif per >=50:
        grade = "D"
    else:
        grade = "F"
    return total,per,grade
```

CamScanner

```
#Input 5 students
for i in range(5):
    name = input("Enter Name")
    roll = input("Enter roll Number")
    p = float(input("Python marks"))
    ai = float(input("AI marks"))
    maths = float(input("Maths marks"))

total, per, grade = result(p, ai, math)

    students.append({
"name": name, "roll": roll, "total": total,
    "percentage": per, "grade": grade
    })
#Display results
    for s in students:
        print(s)
#Highest Percentage
    top = max(students, key=lambda s:
                s["Percentage"])

    print("Highest Percentage:")
        print(top)

#Save results
with open("results.txt", "w") as file:
        for s in students:
            file.write(str(s))
```

```
file.write("Highest Marks")
file.write(str(top))
```

Question # 02

Code

```
def analyze(s):
    w = s.split()
    u = set(w)
    v = sum(c.lower() in for c in s)
    print("Words", len(w))
    print("Characters:", len(s))
    print("Longest:", max(w, key=len))
    print("Shortest:", min(w, key=len))
    print("Vowels:", v)
    print("Unique words:", u)
    print("Alphabetical:", sorted(w))

with open("word-analysis.txt", "w") as f:
    f.write(str([len(w), len(s), max(w,
                key=len),
                min(w, key=len), v, u, sorted(u)
                ]))
s=input("Enter sentence:")
analyze(s)
```

CamScanner

Question #03

Code

```
def bill(products):
    total = 0
    for p in products:
p["total"]=p["price"] * p["quantity"]
    total += p["total"]

discount = total * 0.10 if total >10000
                        else 0

final = total - discount

expensive = max(products, key=lambda p:
                p["price"])

    print ("Total Bill", total)
    print ("Discount", discount)
    print ("Final Bill", final)
    print ("Most Expensive",
                    expensive ["name"])

with open ("bill.txt","w") as f:
    f.write (str (products))
    f.write ("Final Bill :"+str(Final))
```

```
products = []
for i in range(5):
    name = input("Product Name")
    price = float(input("Price"))
    quantity = int(input("Quantity"))
products.append({
    "name": name,
    "price": price,
    "quantity": quantity
})
bill(products)
```

Question #04

Code

```
def add(a,b):
    return a+b
def sub(a,b):
    return a-b
def mul(a,b):
    return a*b
```

CamScanner

```
def mod(a, b):
    return a%b

def div(a , b):
    return a/b

while True:
a=input("Enter 1st Number")
b=input("Enter 2nd Number")
op=input("Enter operation(+,-,*,/,%):")

    if op=="+":
    print(add(a,b))
    elif op=="-":
     print(sub(a,b))
    elif   op==  "*":
     print(mul(a,b))
    elif op=="/":
     print(div(a,b))
    elif op=="%":
     print(mod(a,b))
    else
     print("Invalid operation")
    again=input("Again(Yes/No):")
     if again.lower()!="yes":
          break
```

Question #05

Code

```
menu = {1: ("Burger", 500), 2 : ("Pizza", 800)
        3: ("Fries", 250), 4: ("Drink", 150)}
def order():
    total = 0
while True:
print ("1. Burger 2. Pizza 3. Fries 4. Drink 5. Exit")
    n = int (input ("Choice :"))
    if  n = =0 :
        break
    elif  n in menu :
        q = int (input ("Quantity "))
        total + = menu[n][1] * q
    else:
     print ("Invalid choice")
    print ("Final Bill", total)
    with open("Orders.txt","w") as f :
    f.write ("Final Bill" + str( total))
     order ()
```